Syllabus Details

Syllabus ID:	12556
Syllabus Name:	**Python programming_Lập trình Python**
Course Name English:	**Python programming**
Subject Code:	**PRP201c**
Learning-Teaching Method:	Blended,Online,Inquiry-based Teaching, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 60h online + 3h offline + 1h TE + 2h PE + 84h self-study
Pre-Requisite:	Recommend to have knowledge of basic programming in advance
Description:	This course will introduce fundamental programming concepts including data structures, networked application program interfaces, and databases, using the Python programming language. In the Capstone Project, students'll use the technologies learned throughout the course to design and create your own applications for data retrieval, processing, and visualization.
StudentTasks:	- Student must get the certification of Python for Everybody specialization from Coursera to be accepted to the final examination
Tools:	Internet
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	359/QĐ-ĐHFPT dated 04/09/2025
IsApproved:	**True**
Note:	Assessment scheme: 1. Complete the online course to be allowed to take Final Exam 2. Final Exam is included of Final Theory Exam (TE) & Final Practical Exam (PE): 100% 3. Students who complete all MOOCs before the course deadline will receive 1 bonus point; otherwise, no bonus point will be awarded. 4. Completion Criteria: Final TE Score >=4 & Final PE Score >=4 & FR >= 5 (FR = min(10, (Final TE Score + Final PE Score)/2 + bonus)
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2025

7 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	https://www.coursera.org/learn/python-data-visualization?specialization=python		Coursera				True	False	True	MOOC 5: Capstone: Retrieving, Processing, and Visualizing Data with Python
2	https://learner.coursera.help/hc/en-us/articles/208280036-Coursera-Code-of-Conduct		Coursera				False	False	True	Help Articles

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	https://www.coursera.org/specializations/python		Coursera				True	False	True	SPEC: Python for Everybody Specialization
4	https://www.coursera.org/learn/python?specialization=python		Coursera				True	False	True	MOOC 1: Programming for Everybody (Getting Started with Python)
5	https://www.coursera.org/learn/python-data?specialization=python		Coursera				True	False	True	MOOC 2: Python Data Structures
6	https://www.coursera.org/learn/python-network-data?specialization=python		Coursera				True	False	True	MOOC 3: Using Python to Access Web Data
7	https://www.coursera.org/learn/python-databases?specialization=python		Coursera				True	False	True	MOOC 4: Using Databases with Python

4 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Explain the basics of programming computers using Python
2	CLO2	Understand fundamental programming concepts such as data structures
3	CLO3	Create your own applications for data retrieval and processing
4	CLO4	Describe the basics of the Structured Query Language (SQL) and database design

View mapping of CLOs to PLOs

Download All Student Material 12 sessions (45'/session)

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Specialization Introduction Getting started with Python - Why we Program? - Installing and Using Python - Variables and Expressions Offline 1: Introduction to the online course PRP201c	Offline,Online	LO1				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed) All students should participate in offline slot (if any question)	
2	Getting started with Python - Conditional Code - Functions - Loops and Iteration	Online	LO1				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	Python Data Structures - Strings - Installing and Using Python (for writing Python programs) - Files	Online	LO2				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
4	Python Data Structures - Lists - Dictionaries - Tuples - Graduation	Online	LO2				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
5	Using Python to Access Web Data - Getting Started - Regular Expressions	Online	LO3				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
6	Using Python to Access Web Data - Networks and Sockets - Programs that Surf the Web	Online	LO3				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
7	Using Python to Access Web Data - Web Services and XML - JSON and the REST Architecture	Online	LO3				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
8	Using Databases with Python - Object Oriented Python - Basic Structured Query Language	Online	LO4				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
9	Using Databases with Python - Data Models and Relational SQL - Many-to-Many Relationships in SQL - Databases and Visualization	Online	LO4				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
10	Capstone: Retrieving, Processing, and Visualizing Data with Python	Online	All LOs				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
11	Capstone: Retrieving, Processing, and Visualizing Data with Python	Online	All LOs				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
12	Final Project Peer Review Evaluation Specialization Wrap-Up Offline 2: Review Python for Everybody Specialization	Offline,Online	All LOs				- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed All students should participate in review slot (if any question) Deadline of Coursera completion: End of Friday of the week.	

0 Constructive question(s)

2 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Final Exam (PE)	Final exam	1	50.0%	4	120'			N/A	All studied courses	by Exam Board	Customized from the Assignment & Capstone Project of this specialization
2	Final Exam (TE)	Final exam	1	50.0%	4	60'		Computer gradable	50	All studied courses. Each module of course contributes 3-5 questions.	by Exam Board	Customized from the quizzes of this specialization.